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Forecasting tipping times from data: a practical guideline

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We derive a fully data–drive pipeline for the accurate forceast of tipping times in complex systems by combining established techniques in nonautonomous dynamics and statistical estimation for stochastic differential equations. Our aim is to provide a practical guide for the generation and collection of data to scientists and modelers that are interested in predicting tipping events from recorded timeseries of their systems. This pipeline relies on a series of assumptions, discussed and proved throughout this paper, that form an optimal observation protocol for the forecast of tipping times.

1. Introduction

Tipping phenomena (or critical transitions) are abrupt changes in the state of complex systems that correspond to disruptive and potentially permanent alterations in their long–term behaviour. They have been studied across a broad and diverse collection of natural and human critical phenomena including ecosystems degradation [1,2], neurobiological activity spikes [3], onset of asthma [4,5], weakining of ocean currents [6,7], hysteresis in laser systems [8], generation of instabilities in magnetically confined plasma [9] and thermoacustic systems [10], marine vessels capsizing [11], formation of traffic congenstion [12] and civilization collapse [13,14].

Catastrophic events are usually characterized by the loss of stability of steady–states due to nonlinear and noisy responses to slow–varying external forcing. Their canonical classification, proposed in [15], distinguishes between bifurcation–induced (B-), rate–induced (R-) and noise–induced (N-) tipping points. Furthermore, slow–fast dynamics and bifurcation theory provide a modeling framework for the forecast of critical transitions [16]. The prediction of approaching catastrophes, known as *early-warning signals*, usually relies on the detection of characteristic precursors from non–stationary timeseries.

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The most common of these signatures is known as *critical slowing down* [17] and it is often associated with an increase in the variance and autocorrelation of a timeseries sample.

Several early-warning signals have been proposed in the last two decades; we refer interested readers to [18], and more recently [19], for a concised overview on the state–of–the–art research on the topic. Debates regarding the legitimacy of theoretical early-warnings applied to real–worl data have since followed [20–23], suggesting that caution should be exercised when relying on these indicators to predict critical events [24–27]. In particular, pertinent concerns to applications are:

- recording optimal observables from high–dimensional (spatio–temporal or networked) systems is necessary in order to detect critical slowing down [28];
- reliable and skillfull early-warnings require sufficient timescale separation between the fast (state) and the slow (forcing) variables [16];
- predictions based on ensemble averages without uncertainty quantification of the single realizations can lead to missed warnings (i.e. false negatives) and/or spurious signals (i.e. false positives) [29];
- lack of knowledge of baseline values of classical early-warning signals result in lack of interpretability of statistical quanties without context and historical trends [30].

In this manuscript we are concerned with the problem of forecasting the time at which complex systems will undergo critical transitions given past and present data. The forecast is a (linear) extrapolation of the estimated return rate of nonlinear systems that are slowly ramped towards a saddle-node bifurcation. Such saddle-node B-tipping events are found in many of the applications mentioned above in which the collected timeseries samples provide a discrete time approximation of a slow passage through the bifurcation [31]. This paper provides a data–driven pipeline and an observation protocol with which scientists and modelers can collect data to predict such tipping events, far enough in the future and with sufficient accuracy, to enable policies that stop or reverse the catastrophe.

To achive this we leverage well–known results in classical and nonautonomous dynamical systems theory and stochastic dynamics as well as canonical statistical estimation techniques applied to timeseries data. The pipeline and the protocol rely on a series of assumptions that provide the constraints that the data must satisfy in order for the forecasting technique to yield consistent and reliable results.

The following discussion is structured as follows. In § 2 we introduce the mathematical formalism of tipping times in slow–fast stochastic processes. In § 3 we formulate the extrapolation problem at the core of the data–driven pipeline by presenting three techniques to estimate the return rate of a given sample, two of which have already been proposed in the literature and a new one which we introduce. In § 4 we derive the observation protocoal, that is a series of conditions that allow for the accurate forecast of tipping times from collected measurements or experimental data. In § 5 we apply the proposed methodology to simulated and real–world timeseries data. In § 6 we discuss our findings, address the limitations of the pipeline within the protocol and suggest future directions for this line of research.

2. Tipping time of a slowly ramped systems

The prototypical case of critical transitions in complex system is that of a dynamic saddle-node normal form, with slow (linear) passage through the bifurcation and (additive) noisy fluctuations modelled as an Itô stochastic differential equation of the form

$$\mathrm{d}X_t = f\big(X_t, \mu(t)\big)\,\mathrm{d}t + \sigma\,\mathrm{d}W_t, \qquad f(x,\mu) := -\mu - x^2, \qquad \mu(t) := \mu_0 + \varepsilon t, \tag{2.1}$$

where $X_t \in \mathbb{R}$ is the state of the system, perturbed by a Wiener process W_t with noise amplitude $\sigma > 0$, and $\mu(t)$ is the bifurcation parameter, ramped with constant speed $\varepsilon > 0$ from an initial

value $\mu_0 < 0$. The drift term $f(x,\mu)$ in (2.1) has two equilibria $x_\pm(\mu) = \pm\sqrt{-\mu}$ for any $\mu < 0$; at $\mu = 0$ the two branches collide $x_+(0) = x_-(0) = 0$ in a saddle-node bifurcation and for $\mu > 0$ they annihilate each other. Small perturbations around the stable equilibrium $x_+(\mu)$ decay exponentially fast at a rate dictated by the leading eigenvalue of the linearized vector field

$$\lambda(\mu) := \partial_x f(x_+(\mu),\mu) = -2\sqrt{-\mu} < 0, \quad \mu < 0. \tag{2.2}$$

Specifically, the quantity $1/|\lambda|$ identifies the *correlation* or *relaxation time* of noisy fluctuations towards $x_+(\mu)$. Evidently, $|\lambda| \downarrow 0$ as $\mu \uparrow 0$, which in turn gives asymptotically longer relaxation time and eventually the blow-up $1/|\lambda| \to \infty$. This phenomenon is known as *critical slowing down* and it is a generic behaviour of any complex system driven close enough to a saddle-node bifurcation as the nonlinear dynamics is reduced to the normal form (2.1) by a smooth change in coordinates.

For a given ramp speed $\varepsilon > 0$, the time at which the leading eigenvalue (2.2) vanishes corresponds to the time at which the forcing parameter $\mu(t)$ reaches the saddle-node bifurcation $\mu(t) = 0$, which is $\tau := -\mu_0/\varepsilon > 0$. One can thus rewrite the linear ramp of the forcing parameter to be independent of the initial condition $\mu_0 < 0$ and parametrized by the time horizon τ at which the saddle-node bifurcation is reached, i.e.

$$\mu(t) = \mu_0 + \varepsilon t = \varepsilon(t - \tau) < 0 \quad \forall t < \tau. \tag{2.3}$$

Substituting (2.3) in (2.2) gives an explicit time dependence of the approach to the bifurcation

$$\lambda(t) = -2\sqrt{\varepsilon(\tau - t)} \uparrow 0, \quad t \uparrow \tau. \tag{2.4}$$

(a) Deterministic tipping time

Let us first consider the deterministic limit $\sigma \downarrow 0$ for which the nonautonomous system (2.1) is known as the *dynamic* saddle-node normal form [32]. The time horizon τ (henceforth *breaking time*) identifies the loss of hyperbolicity of the stable equilibrium branch $x_+(t) := x_+(\mu(t))$ as per (2.4). In the nonautonomous regime however, the classic notion of $x_+(t)$ is replaced by the pullback attractor $x_{\mathrm{pb}}(t)$ which is the unique solution of (2.1) (in the absence of noise $\sigma = 0$) satisfying

$$\lim_{t \to -\infty} |x_{\mathrm{pb}}(t) - x_+(t)| = 0. \tag{2.5}$$

For a slow passage through the saddle-node bifurcation, i.e. $0 < \varepsilon \ll 1$, provided that $\mu_0 = \lim_{t\to-\infty} \mu(t) \ll 0$, the pullback attractor converges to a slow manifold [32, Thm. 2.1.7], which tracks the stable equilibrium branch $x_+(\mu(t))$ and persists past the breaking time τ for some time before blowing-up to $-\infty$. The trajectory of a slow passage through a saddle-node bifurcation will thus undergo B-tipping at a delayed time $\tau_{\mathrm{pb}} > \tau$ (henceforth *tipping time*), creating a window of opportunity $\Delta\tau := \tau_{\mathrm{pb}} - \tau > 0$ to reverse the forcing ε and avoid the critical transition.

To quantify the tipping time, and thus identify the window of opportunity, we follow the procedure outlined in [33]. We start by writing the dynamic saddle-node normal form as a Riccati equation

$$\frac{\mathrm{d}x}{\mathrm{d}t} = f(x,\mu(t)) = \sum_{k=0}^{2} c_k(t) x^k, \tag{2.6}$$

with $c_0(t) = \varepsilon(\tau - t)$, $c_1(t) = 0$ and $c_2(t) = -1$, and convert it into a second-order, linear system via change of variables $x \mapsto -(c_2(t)y)^{-1}(\mathrm{d}y/\mathrm{d}t)$

$$\frac{\mathrm{d}^2 y}{\mathrm{d}t^2} = c_0(t) y. \tag{2.7}$$

Then we can rewrite (2.7) in the rescaled time $s := \varepsilon^{1/3}(\tau - t)$ so that it reads as a Airy equation of the form

$$\frac{\mathrm{d}^2 y}{\mathrm{d}s^2} = sy, \tag{2.8}$$

whose general solution is a linear combination of Airy functions of the first $\mathrm{Ai}(s)$ and second kind $\mathrm{Bi}(s)$ [34, § 10.4]. By backward substitution, and in the past limit $t \to -\infty$, the pullback attractor solving (2.6) is given by

$$x_{\mathrm{pb}}(s) = -\varepsilon^{1/3} \frac{1}{\mathrm{Ai}(s)} \frac{\mathrm{d}\,\mathrm{Ai}}{\mathrm{d}s}. \tag{2.9}$$

From (2.9), $x_{\mathrm{pb}} \to -\infty$ as $\mathrm{Ai} \to 0$, which occurs at $s \approx 2.338$. It thus follows that the tipping time of the pullback attractor is given by $\tau_{\mathrm{pb}} \approx \tau + 2.338\,\varepsilon^{-1/3}$ and the opportunity window scales with the reciprocal of the cubic root of the parameter speed, i.e. $\Delta\tau \propto \varepsilon^{-1/3}$.

(b) Tipping time distribution

We now consider the more general case of stochastic fluctuations of slow passage through the bifurcation by restoring $\sigma > 0$ in (2.1). As a result, the tipping time τ_{pb} is treated as a random variable τ_{tip} with an associated probability distribution. To characterize τ_{pb} for (2.1), we follow the procedures outlined in [35,36]. To start, consider again the singular limit $\varepsilon \downarrow 0$ of (2.1), written as a gradient system

$$\mathrm{d}X_t = -V'(X_t, \mu)\,\mathrm{d}t + \sqrt{2D}\,\mathrm{d}W_t, \qquad V(x,\mu) := \mu x + \frac{x^3}{3}, \tag{2.10}$$

where $V(x,\mu)$ is the potential of the vector field $f(x,\mu)$ and $D := \sigma^2/2$. The equilibria $x_\pm(\mu) = \pm\sqrt{-\mu}$ of f correspond to the stationary points of V, respectively its local minimum and local maximum. At any $\mu < 0$, the potential barrier, separating the local minimum $x_+(\mu)$ from the local maximum $x_-(\mu)$ is

$$\Delta V := V\big(x_-(\mu)\big) - V\big(x_+(\mu)\big) = \frac{4}{3}(-\mu)^{3/2} > 0. \tag{2.11}$$

The critical transition of a solution of (2.1), starting in a neighbourhood of $x_+(\mu)$, can thus be associated to the escape of an overdamped particle across the potential barrier ΔV driven by noisy fluctuations of intensity D. For fixed values of $\mu < 0$ such that $\Delta V \gg D$, these escapes are rare events described by Freidlin–Wentzell theory [37] which has been widely applied in many disciplines (see [38] and references therein). In particular, the event of a trajectory X_t of (2.10), starting near $x_+(\mu)$ and reaching $x_-(\mu)$ across the potential barrier ΔV, satisfies a so-called *large deviation principle*. The *first exit time* of such event [39, § 5.5]

$$\tau_{\mathrm{exit}} := \inf\{t \geq 0 \,:\, X_t \leq x_-(\mu)\}, \tag{2.12}$$

is asymptotically exponentially distributed [40, Thm. 2.3]

$$\tau_{\mathrm{exit}} \overset{d}{\approx} \exp\big(k(\mu)\big), \quad D \downarrow 0, \tag{2.13}$$

where the rate parameter is given by Kramers formula

$$k(\mu) = \frac{\sqrt{V''\big(x_+(\mu)\big)\,|V''\big(x_-(\mu)\big)|}}{2\pi} \exp\left(-\frac{\Delta V}{D}\right) = \frac{\sqrt{-\mu}}{\pi} \exp\left(-\frac{4(-\mu)^{3/2}}{3D}\right). \tag{2.14}$$

Notice that the reciprocal of the escape rate in (2.14) gives the *mean first exit time* of (2.12), i.e. $\mathbb{E}[\tau_{\mathrm{exit}}] = 1/k(\mu)$. This result, originally formulated for stationary diffusion [41] and jump processes [42], was then extended to the dynamic saddle-node normal form [35]. Specifically, in the limiting regime of low noise intensity $D \ll \varepsilon/2$, the approximate distribution $\tau_{\mathrm{tip}} \overset{d}{\approx} p_1$ of the tipping time τ_{tip} has been shown to be [35, Eq. (19)]

$$p_1(t) := \frac{\varepsilon^{2/3}}{\sqrt{4\pi^5 D}\;\mathrm{Bi}^2\big(-\mu(t)\varepsilon^{-2/3}\big)\,\mathcal{J}(t)^{1/2}} \exp\left[-\frac{\varepsilon^{2/3}\;\mathrm{Ai}^2\big(-\mu(t)\varepsilon^{-2/3}\big)}{4\pi^2 D\;\mathrm{Bi}^2\big(-\mu(t)\varepsilon^{-2/3}\big)\,\mathcal{J}(t)}\right], \tag{2.15}$$

with $\mathcal{J}(t) := \int_{-\infty}^{t} \mathrm{Ai}^4\big(-\mu(s)\varepsilon^{-2/3}\big)\,\mathrm{d}s$. We remark that $p_1(t)$ has support in the interval $\tau + 1.17\varepsilon^{-1/3} < t < \tau + 3.27\varepsilon^{-1/3}$ and it contains the tipping time of the pullback attractor $\tau_{\mathrm{pb}} \approx$

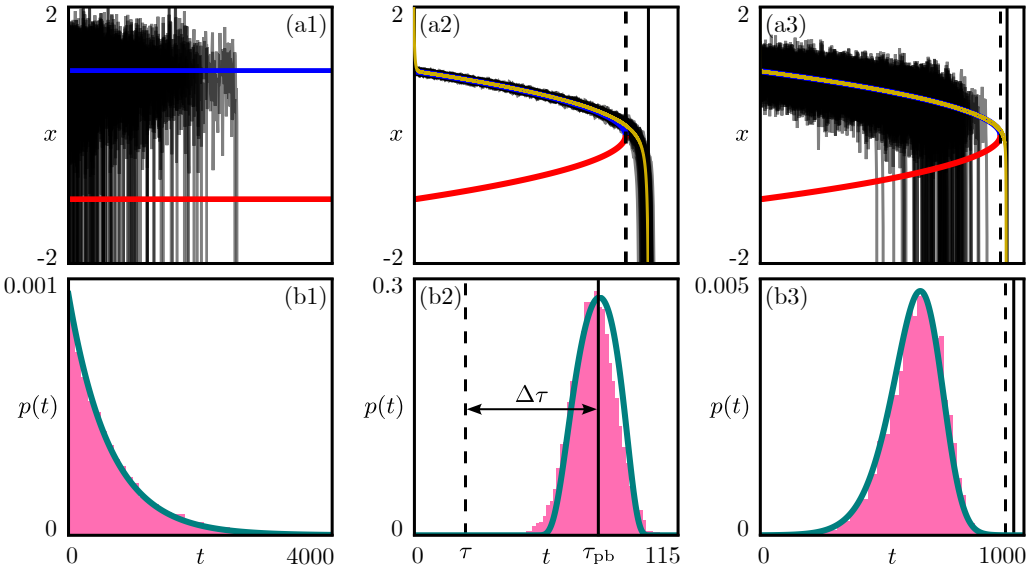


Figure 1. Comparison of the distributions of tipping times of the dynamic saddle–node normal form across different regimes: (a1)–(b1) noise-induced escapes at stationary parameter value $\mu = -1$ and noise level $D = 0.25$; (a2)–(b2) slow–passage escapes at initial parameter value $\mu_0 = -1$, speed $\varepsilon = 10^{-2}$ and noise $D = 10^{-3}$; (a3)–(b3) early escapes at initial parameter value $\mu_0 = -1$, speed $\varepsilon = 10^{-3}$ and noise $D = 0.06$. In panels (a1)–(a3) we show Monte Carlo simulations of an ensemble of $10{,}000$ solutions of (2.1) in each regime, reppresented by the black curves. The blue and red curves reppresent the stable and unstable branches of equilibria $x_{\pm}(\mu(t))$ of f respectively, while the yellow curve depicts the pullback attractor $x_{\mathrm{pb}}(t)$. The vertical dahsed and solid black lines in (a2)–(a3) denote the breaking time τ and tipping time of the pullback attractor τ_{pb} respectively. In panels (b1)–(b3) the empirical distributions of the tipping time τ_{tip} of the Monte Carlo simulations, depicted by the pink histograms, are compared with the analytical approximations (2.13), (2.15) and (2.17) respectively, which are depicted by the teal curves.

$\tau + 2.338\,\varepsilon^{-1/3}$. Furthermore (2.15) is approximately Gaussian locally to its maximum

$$\tau_{\mathrm{tip}} \overset{d}{\approx} \mathcal{N}\Big(\tau_{\mathrm{pb}}\big[1 + O(D/\varepsilon)\big], O\big(D\varepsilon^{-5/3}\big)\Big), \tag{2.16}$$

with mean close to the tipping time of the deterministic solution, i.e. $\mathbb{E}[\tau_{\mathrm{tip}}] \approx \tau_{\mathrm{pb}}$.

In the opposite regime of large noise intensity $\varepsilon/2 \ll D$, and up to $-\mu(t) \gg D^{2/3}$, it was later shown that the approximate tipping time distribution $\tau_{\mathrm{tip}} \overset{d}{\approx} p_2$ has a maximum before the breaking time τ and it has the form [36, Eq. (4)]

$$p_2(t) \coloneqq k\big(\mu(t)\big)\, \exp\left[-\frac{D}{2\pi\varepsilon}\, \exp\left(-\frac{\Delta V\big(\mu(t)\big)}{D}\right)\right], \tag{2.17}$$

where $k(\mu(t))$ is the Kramers escape rate given in (2.14) and $\Delta V(\mu(t))$ is the potential barrier given in (2.11).

The analytical approximations of the three different regimes prescribed in [35,36,40] match the empirical observations obtained via Monte Carlo simulations as reported in Figure 1. Notice how, in the small noise regime $D \ll \varepsilon/2$, almost all the mass of the tipping time distribution (2.15) is located after the breaking time τ and its mean approximates the tipping time of the pullback attractor τ_{pb} (see panel (b2)). This setting thus prescribes the optimal scenario for which the (accurate) forecast of the breaking time τ provides valuable predictions for the estimation of the tipping time since it provides policymakers a window of opportunity $\Delta\tau$ to reverse the forcing of the system towards the saddle–node catastrophe.

3. Forecasting problem

From § 2, we summarize three important results that motivate the forecasting of tipping times:

(i) for a slow passage through a saddle-node bifurcation the leading eigenvalue $\lambda(t)$ explicitly depends on the breaking time τ via Eq. (2.4);
(ii) the time at which a slowly ramped solution tips $\tau_{\rm pb}$ is delayed with respect to the breaking time τ by a factor of $\varepsilon^{-1/3}$ that depends on the speed of the forcing, thereby creating an opportunity window $\Delta\tau$;
(iii) in the small noise regime $\sigma = \sqrt{2D} \ll \sqrt{\varepsilon}$, the delayed tipping time of a noisy solution $\tau_{\rm tip}$ occurs almost always after the breaking time τ, as per the support of the density $p_1(t)$ in Eq. (2.15), and on average close to the deterministic tipping time, as per Eq. (2.16).

Provided a timeseries that is slowly ramped towards a saddle-node bifurcation, with noise σ and speed ε satisftying $0 < \sigma^2 \ll \varepsilon \ll 1$, then, if we accurately forecast the breaking time τ (i), we identify a window of opportunity (ii) and avoid the bifurcation-induced catastrophe with high probability (iii) by stopping or reversing the forcing of the parameter. To formulate the forecasting problem of the breaking time τ we introduce the *(squared) return rate* $\alpha(t) \coloneqq \lambda^2(t)$ which, from (2.4), reads

$$\alpha(t) = 4\varepsilon(\tau - t) > 0 \quad \forall t < \tau. \tag{3.1}$$

Notice that, while $\alpha(t) \downarrow 0$ as $t \uparrow \tau$ just like $-\lambda(t)$, the return rate in (3.1) establishes an affine dependence of the return rate on time t which benefits the forecasting of τ using timeseries data.

To understand this suppose that $\mathcal{X}_1 \coloneqq \{X_j^{(1)} \coloneqq X_{t_j^{(1)}}\}_{j=1,\dots,n}$, $t_j^{(1)} < t_{j+1}^{(1)}$, is a solution of (2.1) that is at safe distance from the bifurcation, i.e. $t_n^{(1)} \ll \tau$. Assume also that we can estimate the true return rate (3.1) from the sample $\mathcal{X}$, which we denote with $\hat{\alpha}_1 \coloneqq \hat{\alpha}(t_n^{(1)})$. We are then presented with another sample $\mathcal{X}_2$, which is also at safe distance from the bifurcation but closer to it than the previous sample $\mathcal{X}_1$, i.e. $t_n^{(1)} < t_n^{(2)}$; once again, assuming we can do so, we estimate the return rate $\hat{\alpha}_2$. Suppose with repeat this procedure for N different samples $\mathcal{X}_1, \dots, \mathcal{X}_N$, yielding a sequence $\hat{\alpha} \coloneqq \{\hat{\alpha}_j\}_{j=1,\dots,N}$. Given this sequence $\hat{\alpha}$ we can then fit a linear model of the form $\hat{\alpha}(t) = \hat{m}t + \hat{q}$, with slope $\hat{m}$ and intercept $\hat{q}$. The forecasted breaking time $\hat{\tau}$ thus corresponds to the zero-crossing of the linear fit, which by extrapolation reads $\hat{\tau} = -\hat{q}/\hat{m}$.

Conseqentially, we translated the problem of forecasting the breaking time τ from timeseries data $\mathcal{X}_1, \dots, \mathcal{X}_N$ into a trivial, linear extrapolation $\hat{\tau}$ from estimated return rates $\hat{\alpha}$. This procedure however features two bottlenecks: first, we need $\hat{\alpha}$ to be a sequence of precise and accurate estimators of the ground truth α and; second, given those estimates $\hat{\alpha}$, we need the linear extrapolation $\hat{\tau} = -\hat{q}/\hat{m}$ to be as close as possible to the ground truth τ to not shrink the window of opportunity. In particular, depending on applications, we would prefer to slightly underestimate the breaking time $\hat{\tau} \lesssim \tau$, rather than slightly overestimating it $\hat{\tau} \gtrsim \tau$, thereby biasing our forecast to minimize missed alarms (i.e. false negatives) in favour of spurious warnings (i.e. false positives). In the remaining of this section we present three data-driven techniques to accurately extract the estimates $\hat{\alpha}$ from a sample $\mathcal{X}$ while we address the extrapolation problem in § 4.

(a) Ornstein-Uhlenbeck approximation

Let us start by characterizing the statistical properties of the noisy fluctuations of a solution $\mathcal{X}$ of (2.1) around the pullback attractor $x_{\rm pb}(t)$ given in Eq. (2.9). We write the solution variable as a superposition $X_t \coloneqq x_{\rm pb}(t) + Z_t$ of a deterministic $x_{\rm pb}(t)$ and a stochastic component Z_t. Substituting this into (2.1), recalling that by definition $\mathrm{d}x_{\rm pb} = f(x_{\rm pb}, \mu)\mathrm{d}t$, we get

$$\mathrm{d}Z_t = \left(-2x_{\rm pb}(t)\, Z_t - Z_t^2\right)\mathrm{d}t + \sigma\, \mathrm{d}W_t. \tag{3.2}$$

At any time $t \ll \tau$ away from the saddle-node bifurcation $f(x_{\mathrm{pb}}, \mu) \approx f\big(x_+(\mu), \mu\big)$ which implies $\partial_x f(x_{\mathrm{pb}}, \mu) \approx \lambda(t)$. Substituting this into (3.2) gives

$$\mathrm{d}Z_t = \big(\lambda(t)\, Z_t - Z_t^2\big)\mathrm{d}t + \sigma\, \mathrm{d}W_t. \tag{3.3}$$

It can be shown that $|Z_t| \approx \sigma/\sqrt{2|\lambda(t)|}$ [16, Thm. 4.1], and thus $Z_t^2 = o\big(|\lambda(t) Z_t|\big)$ for any $\sigma \ll \sqrt{\varepsilon}$ and any $t \ll \tau$ such that $\sigma \ll \sqrt{2}\big(4\varepsilon(\tau - t)\big)^{3/4}$. We can therefore reasonably approximate (3.3) by neglecting the nonlinear term Z_t^2, thereby obtaining

$$\mathrm{d}Z_t \approx \lambda(t)\, Z_t \mathrm{d}t + \sigma\, \mathrm{d}W_t. \tag{3.4}$$

The linear, nonautonomous Eq. (3.4) is a time-inhomogeneous Ornstein-Uhlenbeck process. Suppose the sample $\mathcal{X}$ is recorded across a time window $T_w > 0$ in which $\lambda(t) \approx \lambda$ is approximately constant. Then such sample can be considered as an approximate solution of the stationary process (3.4) which admits a Gaussian distribution [39, § 3.8] with zero mean $\mathbb{E}[Z_t] =: \bar{Z} = 0$ and variance $\mathrm{Var}[Z_t] =: \gamma = \sigma^2/2|\lambda|$. Written in terms of the return rate $\alpha = \lambda^2$, the last identity reads

$$\gamma = \frac{\sigma^2}{2\sqrt{\alpha}}. \tag{3.5}$$

Let $\hat{\gamma}$ be the sample variance of $\mathcal{X}$, then (3.5) gives a sample estimate of the return rate

$$\hat{\alpha} = \frac{\sigma^4}{4\hat{\gamma}^2}. \tag{3.6}$$

The use of the sample variance as an early-warning signal dates back to the very first attempts in predicting bifurcation-induced tipping points and it has since been established as a benchmark precursor of critical slowing down (see e.g. [16] and references therein).

(b) Autoregressive approximation

In the same time window $T_w > 0$ above, the homogeneous Ornstein-Uhlenbeck process has correlation function $\mathrm{Corr}(Y_t, Y_{t+T_w}) = \gamma \exp\big(-\sqrt{\alpha} T_w\big)$ [39, § 3.8], with γ given in (3.5). As a result, the discrete observations in the sample $\mathcal{X}$ can be approximated by fitting a continuous-time autoregressive model of order one

$$X_{j+1} \approx \rho X_j + \sqrt{\gamma(1-\rho^2)}\, \eta_t, \quad \eta_t \sim \mathcal{N}(0,1), \tag{3.7}$$

where

$$\rho := \gamma^{-1}\, \mathrm{Corr}(X_t, X_{t+\Delta}), \quad \Delta = t_{j+1} - t_j, \tag{3.8}$$

is the autocorrelation coefficient at lag-1. Given the sample mean $\hat{X}$ the sample autocorrelation coefficient can be obtained by maximum likelihood estimation [43] and it reads

$$\hat{\rho} = \frac{\sum_{j=1}^{n-1}(X_j - \hat{X})(X_{j+1} - \hat{X})}{\sum_{j=1}^{n}(X_j - \hat{X})^2}. \tag{3.9}$$

Using (3.9) one can estimate the return rate of the sample $\mathcal{X}$ using the autoregressive approximation

$$\hat{\alpha} = \left(\frac{\log \hat{\rho}}{\Delta}\right)^2. \tag{3.10}$$

Autoregressive based early-warning signals have found widespread applications in ecology [44] and climate science [6]. In particular, estimators of the form (3.10) have been used in [45] for estimating the tipping time of the subpolar gyre of the Atlantic Meridional Overturning Circulation system, and in [29] for establishing a framework that quantifies the predicting skill of early-warning signals.

(c) Euler-Maruyama approximation

A solution of the nonlinear system (2.1) in a time window $T_w > 0$ as per above, can be approximated by the process [46, § 8.4]

$$X_{t+1} = X_t + f(X_t, \mu)\Delta + \sigma\sqrt{\Delta}\,\xi_t, \quad \xi_t \sim \mathcal{N}(0,1). \tag{3.11}$$

The sequence above is the Euler-Maruyama approximation of (2.1) and it corresponds to the stochastic generalization of the Euler time-stepping method for ordinary differential equations. We can rearrange the terms in (3.11) to define the first-order time increments of the Euler-Maruyama approximation

$$Y_t := \frac{X_{t+1} - X_t}{\Delta} = f(X_t, \mu) + \tilde{\xi}_t, \quad \tilde{\xi}_t \sim \mathcal{N}\left(0, \frac{\sigma^2}{\Delta}\right). \tag{3.12}$$

Provided a sample $\mathcal{X} = \{X_j\}_{j=1,\dots,n}$ in a time window $T_w = n\Delta$, and a choice of the model function f, Eq. (3.12) can be interpreted as an ordinary least–squares regression problem, with normally distributed residuals $r_j := Y_{j+1} - f(X_j) = \tilde{\xi}_t$. This is part of a broader class of drift inference methods called *quasi-maximum likelihood estimators* [47].

Away from the saddle-node bifurcation, the sample is approximately an Ornstein-Uhlenbeck process of the form (3.4) and thus we can reasonably fit a linear model of the form $f(x;\theta) = \theta x$ to the time increments Y_j. Substituting this ansatz into (3.12), and minimizing the square root of the sum of the squared residuals across the sample, yields the optimization problem

$$\hat{\theta} = \arg\min_{\theta\in\mathbb{R}} \left(\sqrt{\sum_{j=1}^{n-1} \big(Y_j - f(X_j;\theta)\big)^2} \right), \tag{3.13}$$

which is linear in θ due to $f \in \mathbb{P}_1[\theta]$. The solution of (3.13) is given by the normal equation

$$\hat{\theta} = (A^T A)^{-1} A^T Y, \tag{3.14}$$

where $Y = (Y_j)_{j=1,\dots,n-1}$ is the vector of observed time increments of the sample $\mathcal{X}$ and $A = (X_j)_{j=1,\dots,n-1}$ is the model vector of the linear fit f to the sample $\mathcal{X}$. Notice that, by definition, $\hat{\theta}$ is an estimate of the leading eigenvalue λ of the underlying nonlinear evolution and thus

$$\hat{\alpha} = \hat{\theta}^{\,2}, \tag{3.15}$$

is a sample estimate of the return rate using an ordinary (linear) least–squares approximation. The application of quasi-maximum likelihood estimations for the construction of early-warning signals is much more recent than the more established autoregressive and Ornstein-Uhlenbeck approximation, with an early adoption discussed in [30].

4. Observation protocol

The return rate estimators using the Ornstein–Uhlenbeck (3.6), autoregressive (3.10) and Euler–Maruyama (3.15) approximations provide three different, data–drive pipelines to extract $\hat{\alpha}$ from a given sample $\mathcal{X}$ collected over a time window $T_w = n\Delta$, with $n > 0$ number of observations sampled at (regular) frequency Δ. As noted at the beginning of § 3 however, we then use these estimates to forecast the breaking time $\hat{\tau}$ given a sequence of samples $\mathcal{X}_1, \dots, \mathcal{X}_N$. In this section we address this problem in two parts. First, we discuss the accuracy and precision associated to the estimators of the return rate $\hat{\alpha}$ proposed above. Second, we formulate the extrapolation problem of $\hat{\tau}$ given a sequence $\hat{\alpha} = \{\hat{\alpha}_j\}_{j=1,\dots,N}$ of estimators of the return rate and construct the full, data-driven pipeline for the accurate forecasting problem given input timeseries data.

(a) Estimating the return rate

The optimal estimation of the return rate from a finite sample is, perhaps unsurprisingly, highly limited by the quality of the input data provided by the samples $\mathcal{X}_1, \dots, \mathcal{X}_N$. We hereby discuss

three major design constraints that modellers and scientists have to consider when applying the proposed forecasting methodology to their samples.

(i) Admissible time horizon

First observe that the analytical return rate (3.1), on which the sample estimators (3.6), (3.10) and (3.15) are based on, is obtained from the time–dependent eigenvalue (2.4), which is itself evaluated along the drifting quasi–steady equilibrium $x_+(\mu(t))$. However, if the sample $\mathcal{X}$ is assumed to be a slowly forced solution of (2.1) away from the bifurcation, then its trend follows the pullback attractor (2.5) instead of $x_+(\mu(t))$. In the slow regime (after the fast relaxation of the initial condition) the distance at which $x_{\rm pb}(t)$ tracks $x_+(\mu(t))$ increases with time and it reaches its maximum at the saddle-node bifurcation $|x_{\rm pb}(t) - x_+(\mu(t))| = O(\varepsilon^{1/3})$, $t \uparrow \tau$ [32, Thm. 2.2.2]. We are thus interested to quantify a time horizon $t = T_c$ for which the return rate $\alpha(t)$ of the quasi-steady equilibrium is accurate, at least to first-order, to that of the pullback attractor $\alpha_{\rm pb}(t) = \lambda_{\rm pb}^2(t) = 4x_{\rm pb}^2(t)$.

Sufficiently far from the saddle-node bifurcation we can exploit the asymptotic expansion of Airy's function [34, § 10.4] to approximate (2.9) to first-order

$$x_{\rm pb}(s) \approx \varepsilon^{1/3}\left(\sqrt{s} + \frac{1}{4s}\right), \quad s = \varepsilon^{1/3}(\tau - t) \to \infty. \tag{4.1}$$

In this regime, the return rate for the pullback attractor is thus

$$\alpha_{\rm pb}(t) = \alpha(t) + \frac{1}{4(\tau - t)^2} + 2\sqrt{\frac{\varepsilon}{\tau - t}}, \tag{4.2}$$

and dividing each term in (4.2) by $\alpha(t)$ gives

$$\frac{\alpha_{\rm pb(t)}}{\alpha(t)} = 1 + \frac{1}{16\varepsilon(\tau - t)^3} + \frac{1}{2\sqrt{\varepsilon(\tau - t)^3}} = 1 + 2\eta(t) + \eta^2(t) = \big(1 + \eta(t)\big)^2, \tag{4.3}$$

where $\eta(t) \coloneqq 1/(4\sqrt{\varepsilon(\tau - t)^3})$ quantifies the lag correction of the return rate of the pullback attractor.

From (4.3) it follows that $\alpha(t) \approx \alpha_{\rm pb}(t)$ at any time $t \ll \tau$ such that $\eta(t) \ll 1$, from which we obtain the admissible time horizon

$$T_c \lesssim \tau - \varepsilon^{-1/3}. \tag{4.4}$$

The critical threshold (4.4) gives us a criterion for which the estimation of the frozen return rate yields admissible results for the extrapolation of the tipping time, as the lag correction $\eta(t)$ of the pullback attractor with respect to the quasi-steady equilibrium $x_+(\mu(t))$ is negligible. This is depicted in Figure 2 in which a sample path, shown by the grey timeseries in panel (a), follows the pullback attractor solution $x_{\rm pb}(t)$ of (2.1), shown by the yellow curve. Notice how $x_{\rm pb}(t)$ tracks the drifting quasi–steady equilibrium $x_+(\mu(t))$, shown by the blue curve, for some time before the separation becomes visible close to the saddle–node bifurcation. This is reflected in the difference in the return rates of $x_{\rm pb}(t)$ and $x_+(\mu(t))$, depicted by the dashed and solid grey curves in panel (b) respectively. This discrepancy thus identifies an admissible time horizon, shown by the vertical dashed grey curves, bounded below by the exctintion of the transient of the pullback attractor, converging to the (attracting) slow manifold of the nonautonomous system, and above by (4.4).

(ii) Biased estimators

Provided that the sample is in the admissible time horizon $t < T_c$ quantified above, a local estimation of $\alpha(t)$ requires that the sample $\mathcal{X}$ is recorded over a time window T_w in which the leading eigenvalue $\lambda(t)$ at the quasi–steady equilibrium remains approximately constant. This in turn requires the sample to be limited in a finite observation window $T_w = n\Delta$ that induces biases and uncertainties in the estimation of the return rate.

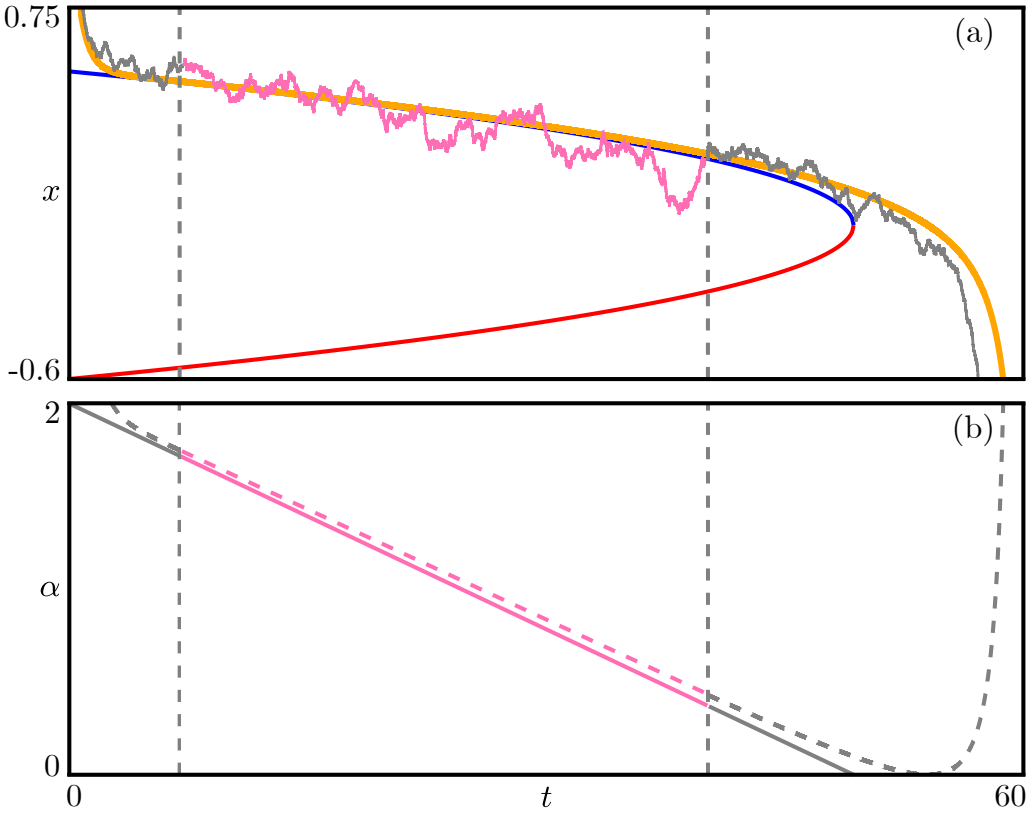


Figure 2. Illustration of the correction in the return rate $\alpha(t)$ between the quasi–steady equilibrium $x_+(t)$ and the pullback attractor $x_{\mathrm{pb}}(t)$. In panel (a) we show a solution of (2.1), given by the grey timeseries, tracking the pullback attractor, depicted by the yellow curve, and eventually separating from the drifting quasi–steady equilibria, given by the blue curve. This solution is generated by setting $\varepsilon = 10^{-2}$ and $\sigma = 0.6\sqrt{\varepsilon}$ in (2.1). In panel (b) we show the effect of this separation on the return rate α; the dashed curve is the return rate evaluate along the pullback attractor, while the solid curve is the return rate evaluated at the quasi–steady equilibrium. The vertical dashed grey lines in both panels reppresent the admissible time horizon for which a subseries of the solution can be used for forecasting, highlighted in pink.

To quantify those, we shall henceforth adopt an ensemble approach in which $\Omega = \{\mathcal{X}_j\}_{j=1,\dots,N}$ denotes a collection of $N > 1$ independent and identically distributed finite samples that correspond to different noise realizations of (2.1). In particular, let us assume that each sample $\mathcal{X}_j$ in the ensemble Ω is a sequence of stationary residuals obtained by detrending a finite windowed solution of (2.1) so that $t_n^{(1)} \equiv t_n^{(N)}$. Then, the resulting Ornstein-Uhlenbeck processes (3.4) can be approximated by order one autoregressive models of the form (3.7).

Then let $\hat{\beta}$ be a generic random variable estimating some ground truth β from a single sample in the ensemble. Denote $\mathbb{E}[\hat{\beta}]$ to be the ensemble mean of $\hat{\beta}$ over Ω and $\mathrm{Var}[\hat{\beta}]$ to be the variance of $\hat{\beta}$ over the ensemble Ω. We thus define the bias $b_{\hat{\beta}}$ and uncertainty $u_{\hat{\beta}}$ of the estimator $\hat{\beta}$ in the canonical fashion

$$b_{\hat{\beta}} \coloneqq \mathbb{E}[\hat{\beta} - \beta] \quad \text{and} \quad u_{\hat{\beta}} \coloneqq \sqrt{\mathrm{Var}[\hat{\beta}]}, \tag{4.5}$$

where, depending on the context, these are also referred to as accuracy and precision respectively; here these two terminologies will be used interchangeably.

Bias in the sample variance The return rate estimator (3.6) uses the sample variance in the Ornstein-Uhlenbeck approximation. It is a standard result that the sample variance $\hat{\gamma} = \mathrm{Var}[\mathcal{X}]$

is a biased estimator for autocorrelated data. Specifically, its accuracy and precision at lag-1 are given by [48, Eq. (4.6)]

$$b_{\hat{\gamma}} = -\frac{2\gamma\rho}{(n-1)(1-\rho)} \quad \text{and} \quad u_{\hat{\gamma}} = \sqrt{\frac{2\gamma^2(1+\rho^2)}{(n-1)(1-\rho^2)}}, \tag{4.6}$$

where ρ is the autocorrelation coefficient given in (3.8). Following a standard delta method [49, § 5.5.4], and using (3.5), we obtain a second-order approximation of the bias in the estimated leading eigenvalue, which reads

$$b_{\hat{\lambda}}^{\text{OUP}} = \frac{|\lambda|}{\gamma} b_{\hat{\gamma}} - \frac{|\lambda|}{\gamma^2} u_{\hat{\gamma}}^2. \tag{4.7}$$

Plugging (4.6) into (4.7) yields a closed form expression for the bias in the estimation of the leading eigenvalue of a finite sample of residuals given its sample variance

$$b_{\hat{\lambda}}^{\text{OUP}} = -\frac{2|\lambda|(1+\rho+2\rho^2)}{(n-1)(1-\rho^2)}. \tag{4.8}$$

Bias in the sample autocorrelation The return rate estimator (3.10) uses the squared logarithm of the sample autocorrelation (3.9), whose bias is quantified in [50, Eq. (21)]

$$b_{\hat{\rho}} = -\frac{1+3\rho}{n-1}.$$

Applying again the delta method, the above yields the bias in the leading eigenvalue [51, Eq. (12)]

$$b_{\hat{\lambda}}^{\text{AC1}} = -\frac{3+\rho^{-2}}{2\Delta(n-1)}. \tag{4.9}$$

Bias in the ordinary least–squares solution The return rate estimator (3.15) is the square of the ordinary least–squares solution of the Euler-Maruyama approximation (3.14), which coincides with the estimated leading eigenvalue of the sample $\mathcal{X}$ by definition. Its bias is thus more straightforward to compute [52, Eq. (4.5)] and it reads

$$b_{\hat{\lambda}}^{\text{OLS}} = \frac{\Delta\lambda^2}{2} - \frac{1+3\rho}{\Delta(n-1)}. \tag{4.10}$$

Notice that, while (4.8) and (4.9) are negative for all samples up to the saddle-node bifurcation $\lambda = -2\sqrt{-\mu} < 0$, the bias in the least–squares estimator (4.10) changes sign at

$$\mu = -\frac{1+3\rho}{2\Delta^2(n-1)}, \tag{4.11}$$

for which (3.15) becomes the only unbiased estimator of the leading eigenvalue.

Accuracy and precision of the estimators Eqs. (4.8)-(4.10) measure the accuracy for the estimation of the leading eigeinvalue λ of a sample. To convert those into biases of the return rate α, notice that

$$\mathbb{E}[\hat{\alpha}] = \mathbb{E}[\hat{\lambda}^2] = \mathbb{E}[\hat{\lambda}]^2 + \text{Var}[\hat{\lambda}] = (b_{\hat{\lambda}} + \lambda)^2 + u_{\hat{\lambda}}^2, \tag{4.12}$$

where in the last equality we used the definitions (4.5). Expanding the quadratic term and using the definition of the return rate α we obtain

$$b_{\hat{\alpha}} = b_{\hat{\lambda}}^2 + 2\lambda b_{\hat{\lambda}} + u_{\hat{\lambda}}^2, \tag{4.13}$$

from which we conclude that the accuracy of the estimators depend on both the accuracy $b_{\hat{\lambda}}$ and the precision $u_{\hat{\lambda}}$ of the leading eigenvalue λ. In the high sampling frequency limit this uncertainty is approximated by the Cramer–Rao inequality [49, Thm. 7.3.9]

$$u_{\hat{\lambda}}^2 \to \frac{2|\lambda|}{\Delta(n-1)}, \quad \Delta \to 0. \tag{4.14}$$

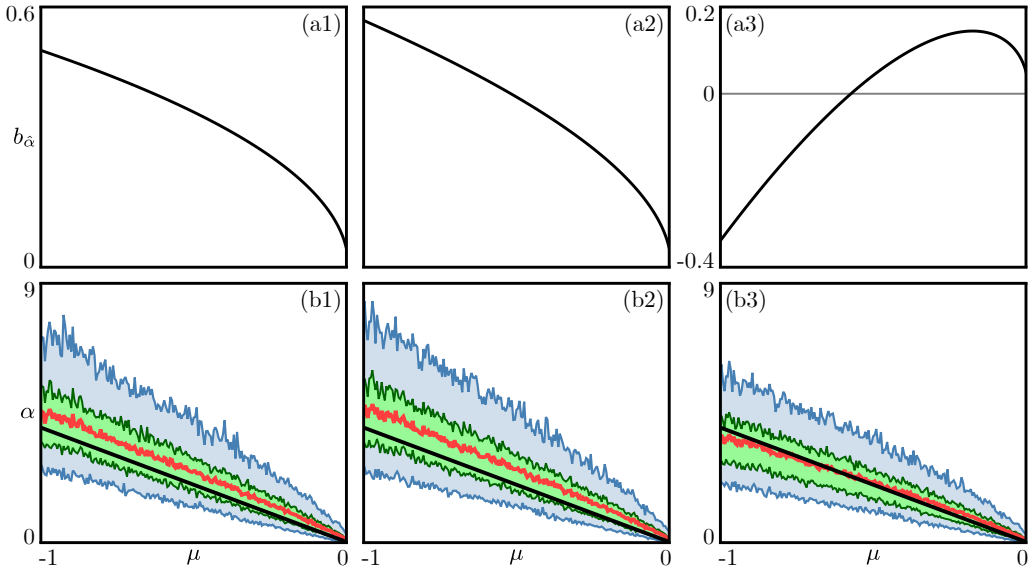


Figure 3. Numerical validation of the return rate bias (4.13) given by: (a1)–(b1) the sample variance estimator (3.6); the autoregressive model fit (3.10) and; the Euler–Maruyama approximation (3.15). In panels (a1)–(a3) we plot (4.13), in the parameter range $\mu \in [-1, 0)$, using (4.8), (4.9) and (4.10) respectively. In panels (b1)–(b3) we plot the results of Monte Carlo simulations of an ensemble $\Omega = \{\mathcal{X}_j\}_{j=1,\dots,N}$ of $N = 200$ independent but stationary solutions of (2.1) with $n = 400$ realizations and noise level $\sigma = 0.02$. At discrete parameter values $\mu \in [-1, 0)$, and for each sample $\mathcal{X}_j \in \Omega$, we estimate $\hat{\alpha}$ using the three techniques discussed above; this forms an ensemble of estimators from which reppresentative statistics are computed. In particular the ensemble mean, shown by the red curves, are compared to the ground truth, black curves, to assess the bias in the estimators; the ensemble interquartile range, shown by the green shaded regions, are used to quantify the uncertainty in the estimators while the blue shaded regions provide the 5th-95th range of the estimators.

Monte Carlo simulations, depicted in Figure 3(b1–b3), show that estimating the return rate of a sample $\mathcal{X}$ using the Euler–Maruyama approach (3.15) yields a more accurate and more precise estimation than the established sample variance (3.6) and autocorrelation (3.10). This result matches the second order approximations for the bias of $\hat{\alpha}$ derived in (4.8), (4.9) and (4.10), which we depict across the parameter interval $\mu[-1, 0)$ in Figure 3(a1–a3).

(iii) Optimal observation length

The topic of selecting an optimal window length to forecast critical transitions was addressed in [53]. In particular, the authors showed that windowed observations with less than 10% of the data from a given sample yield spurious early-warning signals. Such study however only considers fixed window lengths across a solution approaching a critical transition, which neglects the changing characteristic of the solution in the slow passage through the saddle-node. Here we derive the optimal, time-dependent window length of a sample for the accurate forecasting of the tipping time.

Given a sample $\mathcal{X} = \{X_{t_1}, \dots, X_{t_n}\}$ recorded across a window $T_w = t_n - t_1$, we look for bounds of the form

$$L(t) \ll T_{\text{w}} \ll U(t), \tag{4.15}$$

in which the residuals retain the necessary properties for the estimation of the return rate (3.1). The lower bound $L(t)$ is derived by imposing that the sample $\mathcal{X}$ is observed on a window that is

much longer than the relaxation time $1/|\lambda|$ which, from (2.4), reads

$$L(t) = \frac{1}{2\sqrt{\varepsilon(\tau - t)}} \ll T_{\mathrm{w}}. \tag{4.16}$$

This requirement ensures that the realizations X_j in the sample are independent fluctuations around the stable equilibrium $x_+(\mu(t))$. On the other hand, the upper bound $U(t)$ is derived by imposing that the sample $\mathcal{X}$ is quasi-stationary. To derive $U(t)$ observe that, from (2.4), the instantaneous rate of change of the eigenvalue is

$$\frac{\mathrm{d}\lambda}{\mathrm{d}t} =: \dot{\lambda}(t) = \frac{2\varepsilon}{|\lambda(t)|},$$

which is positive and monotonically increasing for any $t < \tau$, which implies that $\dot{\lambda}(t_n) = \sup_{t_1<t<t_n} \dot{\lambda}(t)$ over any finite time window (t_1, t_n). By setting $T_w = t_n - t_1$ it follows

$$0 < \Delta\lambda = \lambda(t_n) - \lambda(t_1) = \int_{t_1}^{t_n} \dot{\lambda}(t)dt < \dot{\lambda}(t_n)T_{\mathrm{w}} = \frac{2\varepsilon}{|\lambda(t_n)|}T_{\mathrm{w}},$$

and dividing each term by $|\lambda(t_n)|$ yields

$$\frac{\Delta\lambda}{|\lambda(t_n)|} < \frac{2\varepsilon}{\lambda^2(t_n)}T_{\mathrm{w}}.$$

Our requirement for the sample $\mathcal{X}$ to be quasi-stationarity across T_w corresponds to small changes of the leading eigenvalue compared to its endpoint value, i.e. $\Delta\lambda/|\lambda(t_n)| \ll 1$, which is guaranteed by imposing

$$\frac{2\varepsilon}{\lambda^2(t_n)}T_{\mathrm{w}} \ll 1,$$

yielding a sufficient condition for the upper bound of the window length

$$T_{\mathrm{w}} \ll U(t) = \frac{\lambda^2(t)}{2\varepsilon} = 2(\tau - t), \tag{4.17}$$

where the last equality follows from (3.1). Plugging (4.16) and (4.17) into (4.15) provides us with the window length protocol

$$\frac{1}{2\sqrt{\varepsilon(\tau - t)}} \ll T_{\mathrm{w}} \ll 2(\tau - t), \tag{4.18}$$

where notice that only the lower bound depends on the speed of the parameter ε.

(b) Extrapolating the tipping time

Having derived the design conditions for the collection of samples that yield optimal estimators $\hat{\alpha}$ we are now ready to specify the full data-driven pipeline for the forecasting of the breaking time τ. As anticipated in § 3, given a sequence of estimators $\hat{\alpha} = \{\hat{\alpha}_j\}_{j=1,\dots,N}$, the forecasting of τ is translated into an extrapolation problem $\hat{\tau} = -\hat{q}/\hat{m}$ via the linear fit $\hat{\alpha}(t) = \hat{m}t + \hat{q}$ of slope $\hat{m}$ and intercept $\hat{q}$. As for the Euler–Maruyama approximation outlined in § 3(c), the linear regression of the sample $\hat{\alpha}$ to the fit $\hat{q}, \hat{m}$ corresponds to a linear least–squares problem whose solution is given by the normal equation

$$\begin{pmatrix} \hat{q} \\ \hat{m} \end{pmatrix} = \left(B^T B\right)^{-1} B^T \hat{\alpha}, \tag{4.19}$$

where $B = (1, t_j)_{j=1,\dots,N}$ is the model matrix and $\hat{\alpha}_j = \hat{\alpha}(t_j)$ is the estimated return rate at time t_j. To be an efficient estimator of $(\hat{q}, \hat{m})^T$ the solution (4.19) requires that the uncertainty in the input data is homogeneous (i.e. assuming *homoskedasticity*) . This however is clearly not the case for the return rate sample $\hat{\alpha}$ as evidenced in Figure 3(b1–b3); in particular, all three estimators (3.6), (3.10) and (3.15) appear to have decreasing uncertainty as the system is driven closer to the saddle-node bifurcation. Therefore the heterogeneity in the uncertainty of the sample $\hat{\alpha}(t)$ (i.e. *heteroskedasticiy*) makes the ordinary least–square regression (4.19) an inefficient point estimate of

the linear trend $\hat{m}$ and intercept $\hat{q}$ [54]. Assuming that, albeit heterogeneous, the uncertainty $u_{\hat{\alpha}_j}$ in the estimate $\hat{\alpha}_j$ is uncorrelated with that of any other estimate $\hat{\alpha}_k$ in the sequence $\hat{\alpha}$, then the weighted least–squares formulation

$$\begin{pmatrix} \hat{q} \\ \hat{m} \end{pmatrix} = \left(B^T W B\right)^{-1} B^T W \hat{\alpha}, \quad W = \operatorname{diag}\left(u_{\hat{\alpha}_j}^{-2}\right)_{j=1,\dots,N}, \tag{4.20}$$

provides the best possible (unbiased) estimator for the linear fit [55]. Using the squared residuals of the ordinary least–squares solution (4.19) as heteroskedasticiy–consistent estimators of the entries of the weight matrix W [54, Thm. 1], one can thus implement the full, data-driven (weighted) extrapolation problem (4.20), from the sequence $\hat{\alpha}$, to forecast the breaking time $\hat{\tau}$.

5. Practical applications

The tipping time framework and observation protocal, specified in § 2 and 4 respectively, provide us with a series of constraints to collect and/or generate timeseries timeseries samples in order to apply the proposed, data–driven, forecasting pipeline. These constraints allow for the optimal design of experiments, data collection and modeling choices for target applications of slowly passage through saddle-node B-tipping catastrophes. These specifications are summarized as follows:

(i) the noisy fluctuations in the residuals $\mathcal{X}$ are additive, i.e. $\sigma(x) = \sigma > 0$, and small compared to the timescale separation between the state x and its parameter μ, i.e. $\sigma \ll \sqrt{\varepsilon}$;
(ii) the residuals are collected prior to the critical transition and within the admissible time horizon $T_c < \tau - \varepsilon^{-1/3}$;
(iii) the residuals are sampled at regular frequency resulting in constant, small enough time intervals Δ to control bias and uncertainty in the estimation of the return rate $\hat{\alpha}$;
(iv) the residuals are collected over a time window that is long enough so that the correlation in the observation is negligible, i.e. $1/(2\sqrt{\varepsilon(\tau - t)}) \ll T_w$, and short enough so that the sample can be considered a stationary approximation of the nonlinear evolution, i.e. $T_w \ll 2(\tau - t)$.

In this section we apply the above to design the optimal conditions for which data can be collected to accurately forecast the breaking time of the underlying system.

To do so we consider both low- and high-dimensional cases. First we apply our pipeline to the canonical dynamic saddle-node normal form (2.1) to show how the observation protocol can be used to the design of data collection from a simplified model. We then subsequentially consider a one–dimensional observable reppresenting the stength of volume transport in the thermohaline ocean circulation generated by a high–dimensional Community Earth System Model (CESM).

(a) One–dimensional tipping in the dynamic saddle-node normal form

We first consider the low–dimensional case of the dynamic saddle–node normal form perturbed by additive noise as in (2.1). To show how the observation protocol, discussed in § 4, can be exploited to maximize the forecasting power of the proposed pipeline, we perform numerical simulations that guarantee the small passage through the bifurcation and delayed tipping in the small noise regime as shown in Figure 1(a2–b2).

The results of this simulation are depicted in Figure 4. We fix the initial condition at the stable branch $x_+(\mu_0)$ of the equilibria of f (blue curve in panel (a1)) for the initial value $\mu_0 = -1$ of the bifurcation parameter. The parameter speed is set at $\varepsilon = 5 \cdot 10^{-4}$ and, from (2.3), the ground truth of breaking time is $\tau = 2000$ (black dashed vertical line in (a1)). The tipping time of the pullback attractor (black solid vertical line in (a1)) reads $\tau_{\mathrm{pb}} \approx \tau + 2.338\varepsilon^{-1/3} \approx 2029.5$, which identifies a short window of opportunity $\Delta\tau \approx 29.5$. The noise intensity is set to $\sigma = 10^{-2}$ which satisfies the small noise regime approximation (2.15); this in turn guarantees that the tipping time

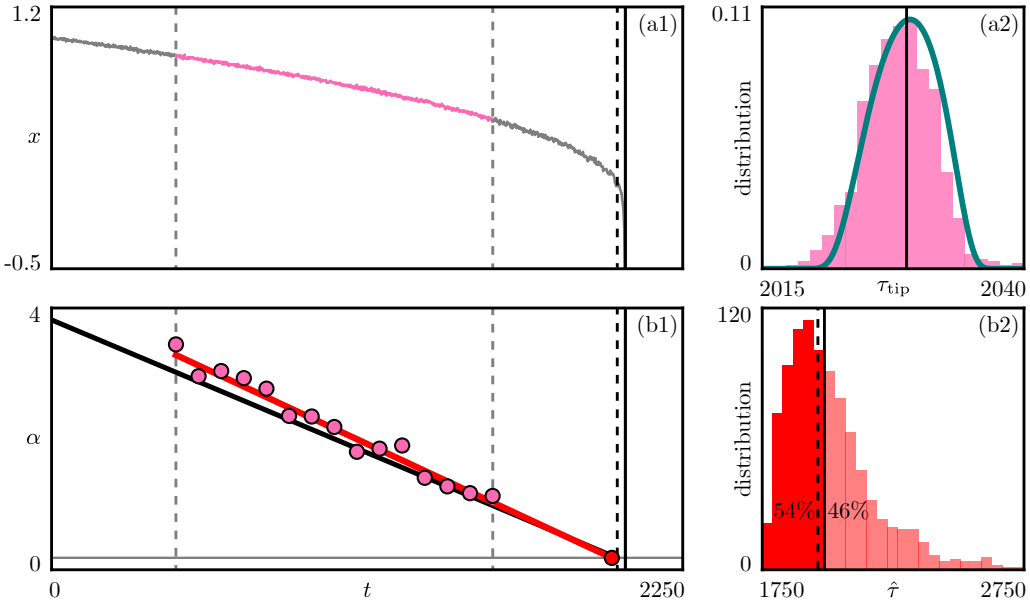


Figure 4. Application of the forecast pipeline to the dynamic saddle–node normal form (2.1). An ensemble Ω of $N = 1000$ sample paths is simulated from $\mu_0 = -1$ with additive noise level $\sigma = 10^{-2}$. Each solution, of which a reppresentative trajectory is show by the grey curve in panel (a1), is slowly driven through the saddle–node bifurcation at time $\tau = 2000$, indicated by the black dashed vertical line in panels (a1) and (b1)-(b2), with ramp speed $\varepsilon = 5 \cdot 10^{-4}$. The tipping distribution, shown by the pink histogram in panel (a2), follows the small noise approximation $p_1(t)$ given in (2.16), shown by the teal curve in panel (a2), and its mean approximates the deterministic tipping time τ_{tip} of the pullback attractor, depicted by the black solid vertical line in all panels. The application of the forecasting pipeline is then applied to the pink subseries in panel (a1), which is extracted from the sample path within the time window $t \in [400, 1600]$, depicted by the grey dashed vertical lines in panel (b1), according to the admissible time horizon (4.4). The estimated return rates $\hat{\alpha}$, depicted by the pink circles in panel (b1), are computed across strides of a rolling window by solving the ordinary least–squares regression in (3.13). The forecast of the breaking time $\hat{\tau}$, reppresented as the red circle in panel (b1), is then obtained by extrapolation, solving the weighted least–square problem (4.20) as shown by the red linear fit in panel (b1). The empirical distribution of forecasted breaking time is reported in panel (b2) with approximately 54% of its mass predicting a breaking time earlier than the deterministic tipping time, highlighted in bright red.

distribution of an ensemble of simulated sample paths (pink histogram in (a2)) is approximately Gaussian (2.16) (teal curve in (a2)) and the tipping time random variable τ_{tip} attains values almost always larger than breaking time τ.

We simulate an ensemble $\Omega = \{\mathcal{X}_k\}_{k=1,\dots,N}$ of $N = 1000$ independent sample paths (grey curve in panel (a1)); each realization $X_{t_j}^{(k)} \in \mathcal{X}_k$ is sampled at regular frequency $\Delta = 10^{-2}$. According to the observation protocol § 4(a), from each sample path we extract a subseries (pink curve in panel (a1)) corresponding to the time window $t \in [400, 1600]$ (grey dahsed vertical lines in panel (a1)), where the lower bound guarantees the extintion of the transient of the initial condition and the upper bound is given by the admissible time horizon (4.4). Each subseries is then analysed in a rolling window technique to estimate the return rate; from (4.18) we set the window length to be $T_{\text{w}} = 80$ which correpsonds to $n = 8000$ observations for each sample.

For each stride of the rolling window we then use the sample to estimate the return rate (3.1) (black curve in panel (b1)) of the underlying, quasi–stationary residuals, which we obtain via linear detrending. Using the pipeline derive in § 3, we apply the Euler–Maruyama approximation and solve the ordinary least–squares problem (3.13) obtaining the least biased estimator (see Eq. (4.10) and Figure 3) of the three methods we discussed above. After a number of strides we obtain a sequence $\hat{\alpha}$ (pink circles in panel (b1)) of estimated return rates of varying precision

(homoskedasticity assumption). By § 4(b), we fit a linear model (red curve in panel (b1)) to $\hat{\alpha}$ by solving the weighted least–squares problem (4.20) which finally yields the forecast $\hat{\tau}$ (red circle in panel (b1)).

By repeating the same procedure for any sample path $\mathcal{X}_k$ in the ensemble Ω we can construct an empirical distribution of the estimated breaking time $\hat{\tau}$, which is shown by the red histogram in panel (b2). In this distribution, we highlight those estimates that forecast a breaking time $\hat{\tau}$ smaller than the tipping time of the pullback attractor τ_{pb}, which approximates the mean of the ground truth distribution $p_1(t)$ given in (2.16), indicated by the black solid vertical line. The total mass of this distribution restricted to $(-\infty, \tau_{\mathrm{pb}})$ is approximately 54% of the total number of estimated breaking times. This entails that, within the simplest model, following an optimal protocal for the generation of data and minimizing the bias in the estimators, the pipeline can still forecast breaking times outside the window of opportunity $\Delta\tau$ with large probability, which might result in delayed or missed warnings.

(b) Collapse in the circulation stength of the North–Atlantic Ocean

The Atlantic Meridional Overturnic Circulation (AMOC) is a volume transport system driving the surface circulation of currents in the Atlantic Ocean. Due to the observed, sustained weakening of its strength, such circulation system has been a major focus of studies in climate science and oceanography communities for the past two decades. It is hypothesized that the collapse of the AMOC, due to instability-induced transitions, can drastically alter the global climate with particularly catastrophic effects in North–Western Europe [7].

The mathematical modeling of the AMOC weakening stength has followed a low-fidelity, conceptual approach, in which the fundamental dynamics is captured by simplified box–models, and a high-fidelity, computational approach, in which large–scale simulations discretise the North–Atlantic ocean in a relatively fine grid and solve for the governing equations of motions. Among the former group, we highlight the canonical $2-$box model proposed by Stommel in 1961 [56], the more recent $5-$box extension introduced by Wood et al. in 2019 [57] and the convective, feedback–delayed multidecadal model of Falkena et al. in 2021 [58]; saddle-node B-tipping and R-tipping have been found in all of the above mentioned models [59–61]. On the other hand, large–scale, multiphysics simulations, such as those provided by Global Climate Models [62], are usually too complex to analyze in a systematic and rigorous way; nonetheless saddle-node B-tipping have been hypothesized [63] and early-warning signals have been proposed [64,65] providing supporting evidence to the conceptual models that the AMOC indeed currently on course toward a collapsed state.

In this section we apply the data–driven pipeline introduced in this work to predict the breaking time of the AMOC using large–scale simulations. Specifically we use the circulation strength timeseries of the fifth version of the Coupled Model Intercomparison Project (CMIP5) in the Community Earth System Model (CESM) as provided in [64].

6. Conclusions

So far the theory of early-warning signals has largely been developed within the same setup we considered through this paper: slow, linear passage through saddle-node bifurcations in low-dimensiona, additive noisy systems. This leaves a large set of models for which the theory has not been developed yet, if not in very incremental steps. The relaxation of the above assumptions requires extensive mathematical development; recent early development being made in that direction include works on systems driven by multiplicative noise [66] and jump processes [67], nonlinear [68] and periodically [69] forced parameter shifts with no timescale separattion, and spatially extended models [70–73].

Data Accessibility. The code used to generate the figures is available at the repository https://github.com/papadeiv/Research/tree/master/2027_proc_r_soc_a.

Authors' Contributions. D.P.: Conceptualization, Methodology, Formal analysis, Investigation, Writing - Original draft preparation; L.S.: Supervision, Conceptualization, Validation, Writing - Review & editing; G.D.: Project administration, Supervision, Funding acquisition, Conceptualization, Validation, Writing - Review & editing.
All authors gave final approval for publication and agreed to be held accountable for the work performed therein.

Conflict of interests. We declare we have no competing interests.

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Appendix A

(a) Something important

Example table.

Table 1. An Example of a Table

date	Dutch policy	date	European policy
1988	Memorandum Prevention	1985	European Directive (85/339)
1991–1997	**Packaging Covenant I**		
1994	Law Environmental Management	1994	European Directive (94/62)
1997	Agreement Packaging and Packaging Waste		

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